1. **Course number and name**
CHE 397: Chemical Engineering Design II

2. **Credits and contact hours**
Credit hours: 4. Contact hours: Two, 75 min lecture periods; Engineering.

3. **Instructor's or course coordinator's name**
Betul Bilgin

4. **Textbook, title, author, publisher, year**
Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of
Plant and Process Design, Second Edition, Elsevier, 2012.
   a. Other supplemental materials
   Seider, Seader, Lewin, Widagdo, Product and Process Design Principles, 3rd Edition, Wiley, 2009
   Writing Style and Standards in Undergraduate Reports, 3e, by Jeffrey Donnell, Sheldon Jeter, Colin MacDougall, and Jacqueline Snedeker

5. **Specific course information**
   a. Brief description of the content of the course (catalog description)
   Application of principles and design methodology of chemical engineering to the design of large-scale chemical processes and plants. A major design project is assigned for solution and presentation at the College Engineering Expo by students working in small groups with industry mentors. Extensive computer use required.
   b. Prerequisites or co-requisites
   CHE 396
   c. Indicate whether a required, elective, or selected elective.
   Required

6. **Specific goals for the course**
   a. Specific outcomes of instruction.
   This course involves the application of all prior CHE concepts to the chemical process design. Design needs to be analyzed from safety, environmental impact, feasibility, risk and economics perspectives. Course aims to equip students with the skill to evaluate open-ended design problems conceptually, analytically and practically.

Students will be able to:

- Develop design plan and project charter for each design phase
- Work effectively with team members
- Present effectively technical material based on audience
- Identify key components in market analysis and construct design basis
- Analyze competing processes

- Write technical content in a way that is appropriate to the audience
- Apply economic analysis
- Identify potential safety concerns in the process and formulate strategies to mitigate those
- Simulate full process with Aspen Plus or SuperPro
- Size process equipment

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3) an ability to communicate effectively with a range of audiences.				X
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.				X
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.			X	
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. Brief list of topics to be covered

- Process description, Design basis, BFD, PFD Flowsheet development
- Equipment list and sizing
- Utilities demand summary for entire plant
- Process simulation
- Process control strategy, process parameters
- Materials of construction
- Plant Cost Analysis (NPV, sensitivity analysis, payback period, capital cost, Aspen economic analysis
- Environmental and safety concerns, process hazards, mitigation.